6. **Further matrix algebra**		
6.1	Linear transformations of column vectors in two and three dimensions and their matrix representation.	Extension of work from FP1 to 3 dimensions.
6.2	Combination of transformations. Products of matrices.	The transformation represented by **AB** is the transformation represented by **B** followed by the transformation represented by **A**.
6.3	Transpose of a matrix.	Use of the relation $(\mathbf{AB})^{\mathrm{T}} = \mathbf{B}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}$.
6.4	Evaluation of 3×3 determinants.	Singular and non-singular matrices.
6.5	Inverse of 3×3 matrices.	Use of the relation $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.
6.6	The inverse (when it exists) of a given transformation or combination of transformations.	
6.7	Eigenvalues and eigenvectors of 2×2 and 3×3 matrices.	Normalised vectors may be required.
6.8	Reduction of symmetric matrices to diagonal form.	Students should be able to find an orthogonal matrix **P** such that $\mathbf{P}^{\mathrm{T}}\mathbf{AP}$ is diagonal.